State-Dependent Contraction Bounds for Reinforcement Learning-Based Nonlinear Control

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***Abstract*—While proving uniform contraction bounds is now a tractable convex optimization problem, analyzing state-dependent rates to establish tighter contraction bounds remains a long-standing challenge. In this work, we provide the conditions on system dynamics necessary for the existence of state-dependent contraction rates. We propose a theoretical reinforcement learning (RL) framework that offers a natural pathway to leverage higher contraction rates using its optimality. Specifically, for RL algorithms designed to minimize cumulative tracking errors defined by uniform-rate contraction metrics, we formally establish state-dependent contraction bounds that decay at the highest feasible contraction rate at each state.**

***Index Terms*—Article submission, IEEE, IEEEtran, journal, LaTeX, paper, template, typesetting.**

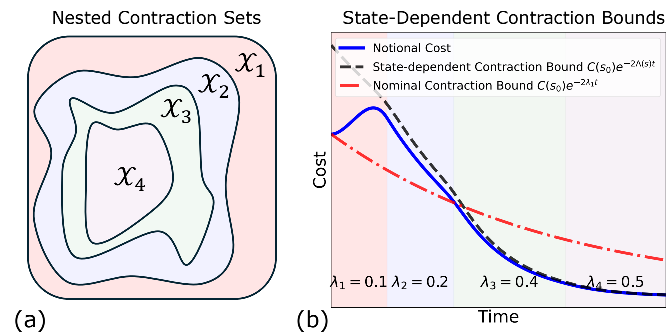

Fig. 1: (a) Nested contraction sets, where each successive subset guarantees a higher contraction rate. (b) An illustrative error (cost) plot showing how state-dependent contraction takes place across the subsets in (a). It highlights that progressively faster contraction rates become available as the system transitions to state subsets where higher contraction is available.

I. Introduction

The existence of contraction metrics [1]—Riemannian metrics under which any two trajectories of a dynamical system converge exponentially—certifies incremental exponential stability (IES). For control-affine systems, the contraction metrics are used to synthesize control laws that render the closed-loop systems contracting [2]. Such synthesis, however, is often conservative, as the contraction condition is imposed with a single rate that must hold uniformly across the state space. Thus, the resulting controller is limited when a subset of states allows for higher contraction rates, as it only certifies a much lower global rate.

We show that reinforcement learning (RL)—minimizing the cumulative tracking error defined by a contraction metric—resolves this conservatism through its optimality. Specifically, we propose state-dependent contraction bounds of optimal policies trained by the RL algorithms and show the bounds decay at the highest feasible rate at that state, while the optimal policies were optimized using contraction metrics that certify a slower uniform rate. However, we note that this benefit comes at a cost, as it requires a larger overshoot margin.

To establish this, we proceed in two steps. First, we characterize when a state-dependent contraction rate exists—that is, when the highest certifiable rate genuinely varies across the state space rather than remaining fixed. Second, for systems in this regime, we derive state-dependent contraction bounds for RL-optimized policies: at each state, the tracking error decays at the highest feasible rate for that state, even though the underlying metric is certified with only a uniform contraction rate. In the worst case, the bound reduces gracefully to that uniform rate.

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II. Preliminaries

A. Dynamical System and Contraction Metrics

In this work, we consider control-affine nonlinear dynamical systems of the form:

$$\dot{x} = f(x) + B(x)u, \tag{1}$$

where $x \in \mathcal{X} \subset \mathbb{R}^n$ is the n-dimensional state, $u \in \mathcal{U} \subset \mathbb{R}^m$ is the m-dimensional control input, $f : \mathcal{X} \to \mathbb{R}^n$ represents the drift dynamics, and $B : \mathcal{X} \to \mathbb{R}^n \times \mathbb{R}^n$ is the actuation matrix function.

Solution trajectories of Equation (1) contract toward one another if there exists a uniformly bounded, positive-definite contraction metric $M(x)$ satisfying $\underline{m}I \preceq M(x) \preceq \overline{m}I$ for $0 < \underline{m} \leq \overline{m} < \infty$, along with the following condition:

$$\dot{M} + \widehat{M(A+BK)} + 2\lambda M \preceq 0, \tag{2}$$

where the arguments are omitted for brevity. Here, $\lambda > 0$ defines the contraction rate, and the hat operator denotes the symmetric sum ($\widehat{A} := A + A^\top$). The matrix $A(x,u) = \frac{\partial f}{\partial x} + \sum_{i=1}^m u_i \frac{\partial b_i}{\partial x}$ is the Jacobian of the system dynamics, and $K(x,t) = \frac{\partial \pi_c}{\partial x}$ represents the differential feedback gain of the baseline policy.

B. Reinforcement Learning

We formulate the nonlinear control problem as a discounted Markov decision process (MDP) [3], defined by the tuple $(\mathcal{S}, \mathcal{U}, P, R, \gamma)$, where $\mathcal{S}$ and $\mathcal{U}$ denote the state and control spaces, respectively. The environment dynamics are characterized by the transition probability density $P(s' \mid s, u) : \mathcal{S} \times \mathcal{U} \times \mathcal{S} \to [0, \infty)$, while $R(s, u, s') : \mathcal{S} \times \mathcal{U} \times \mathcal{S} \to [0, R_{\max}]$ represents the reward function, and $\gamma \in [0, 1)$ is the discount factor. The value function $V_R^\pi(s)$ and the overall performance $J_R(\pi)$ under policy π are defined as:

$$V_R^\pi(s) := \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R(s_{t+k}, u_{t+k}) \mid s_t = s\right], \tag{3}$$

$$J_R(\pi) := \mathbb{E}_{s_0\sim d_0}\left[V_R^\pi(s_0)\right], \tag{4}$$

where d_0 denotes the initial state distribution. The deterministic optimal policy $\pi^\star$ is then given by:

$$\pi^\star := \operatorname*{argmax}_\pi J_R(\pi). \tag{5}$$

III. State-Dependent Contraction Analysis

We characterize when a control-affine system admits a state-dependent contraction rate and reduce this question to a single, checkable condition on the pair $(A(x), B(x))$, where $A(x) := \partial f/\partial x$.

For a contraction-feasible system, let the per-state optimal certified rate be defined as follows:

$$\lambda^*(x) := \sup\{\lambda : \mathcal{P}(\{x\}) \text{ is feasible}\}. \tag{6}$$

Definition 1 (Contraction-Rate Classes)**.** *The dynamics are* uniform-rate *(Class I) if $\lambda^*(\cdot)$ is constant on $\mathcal{X}$, and* state-dependent *(Class II) if $\lambda^*(x_1) \neq \lambda^*(x_2)$ for some $x_1, x_2 \in \mathcal{X}$.*

The certification problem $\mathcal{P}(\{x\})$ is invariant under the orthogonal congruence $\bar{W} \mapsto T\bar{W}T^\top$, $T \in O(n)$. Hence $\lambda^*(x)$ depends on $(A(x), B(x))$ only through its orthogonal-equivalence class, which yields the following test.

Proposition 1 (Orthogonal Reduction Test)**.** *If there exist a map $T : \mathcal{X} \to O(n)$ and a fixed pair (A_0, B_0) such that*

$$T(x)A(x)T(x)^\top = A_0,\ T(x)B(x)B(x)^\top T(x)^\top = B_0B_0^\top \tag{7}$$

for all $x \in \mathcal{X}$, then $\lambda^*(x) = \lambda_0^*$ for all $x \in \mathcal{X}$, where λ_0^* is the optimal certified rate of (A_0, B_0); the rate is uniform. Consequently, a state-dependent rate can arise only when no such reduction exists.

Proof Sketch. Under (7), the congruence $\bar{W} \mapsto T(x)\bar{W}T(x)^\top$ (with $T(x)^\top T(x) = I$) maps $\mathcal{P}(\{x\})$ onto the fixed problem for (A_0, B_0); orthogonal invariance of the underlying condition then gives identical feasibility at every x, so $\lambda^*(\cdot) := \lambda_0^*$. □

Proposition 1 acts as a structural screen. If the pairs reduce to a common (A_0, B_0), the system is uniform-rate and no per-state analysis is needed; if the reduction fails, the certified rate may vary over the state space.

A. Discovery of the contraction rate

We now describe how $\lambda^*(\cdot)$ is computed in practice for the control-affine system in Equation (1), with drift Jacobian $A(x) := \partial f/\partial x$. Because certifying the contraction condition at every point of the continuous set $\mathcal{X}$ is generally intractable, we enforce it on a finite sample

$$\mathcal{E} = \{x_k\}_{k=1}^N \subset \mathcal{X}, \qquad x_k \sim \mathrm{Unif}(\mathcal{X}), \tag{8}$$

and introduce a margin $\varepsilon \geq 0$ that, under appropriate smoothness conditions, extends the certificate to a neighborhood of each sample. For a prescribed rate λ, we obtain the largest-margin contraction metric over $\mathcal{E}$ using the ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) framework [4]:

$$\min_{\bar{W}_k,\,\nu,\,\chi} \quad c_0\chi + c_1\nu \tag{9}$$

$$\text{s.t.} \quad I \preceq \bar{W}_k \preceq \chi I, \quad \forall k \in \{1, \dots, N\}, \tag{10}$$

$$S_k \preceq -\varepsilon I, \quad \forall k \in \{1, \dots, N\}, \tag{11}$$

$$\nu > 0, \quad \nu \leq \underline{w}^{-1}, \quad \chi \leq \nu\overline{w}, \tag{12}$$

where c_0 penalizes the steady-state tracking error, c_1 penalizes the feedback gain, $A_k := A(x_k)$, $B_k := B(x_k)$, and

$$S_k := \dot{\bar{W}}_k + A_k\bar{W}_k + \bar{W}_kA_k^\top + 2\lambda\bar{W}_k - \frac{2\nu}{r}B_kB_k^\top. \tag{13}$$

The set-level rate $\lambda^*(\mathcal{E})$ is then recovered by maximizing λ over the feasibility of this program, e.g., by bisection. The deployed metric follows by undoing the normalization $\bar{W} = \nu W$:

$$M(x_k) = W(x_k)^{-1} = \left(\frac{\bar{W}_k}{\nu}\right)^{-1} = \nu\,\bar{W}_k^{-1}. \tag{14}$$

B. Examples of State-Dependent Contraction Rates

We present control-affine systems where the certified contraction rate is inherently state-dependent, and identify the structural properties that induce this behavior. Both examples utilize the state vector $x = (p, \theta, v, \omega)$—representing position, pitch, body velocity, and pitch rate, respectively—feature a single control input, and are certified over the domain $|\theta| \leq \pi/3$.

a) CartPole: Let m_c denote the cart mass, m_p the pole mass, l the pole length, and g the gravitational acceleration. Defining $\Delta(\theta) := m_c + m_p\sin^2\theta$, the system dynamics $f(x)$ and input matrix $B(x)$ are given by:

$$f(x) = \begin{bmatrix} v \\ \omega \\ \dfrac{m_p\sin\theta\left(l\omega^2 - g\cos\theta\right)}{\Delta(\theta)} \\ \dfrac{m_pl\omega^2\cos\theta\sin\theta - (m_c+m_p)g\sin\theta}{l\,\Delta(\theta)} \end{bmatrix}, \tag{15}$$

$$B(x) = \begin{bmatrix} 0 & 0 & \dfrac{1}{\Delta(\theta)} & \dfrac{\cos\theta}{l\,\Delta(\theta)} \end{bmatrix}^\top.$$

b) Segway: Utilizing the linearized-actuator model, and maintaining the same state ordering, the dynamics are ex-

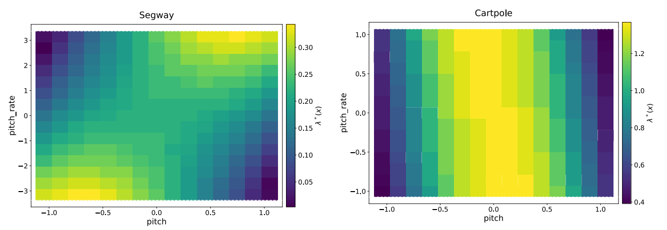

Fig. 2: Certified rate $\lambda^*(x)$ over the each state subsets (grids). The CartPole and Segway systems exhibit varying control authority ($\mathrm{tr}(BB^\top)$), confirming state-dependent optimal rates that vary by states.

pressed as:

$$f(x) = \begin{bmatrix} v \\ \omega \\ \dfrac{\cos\theta\,(9.8\sin\theta + 11.5v) + 68.4v - 1.2\,\omega^2\sin\theta}{\cos\theta - 24.7} \\ \dfrac{-58.8\,v\cos\theta - 243.5\,v - \sin\theta\,(208.3 + \omega^2\cos\theta)}{\cos^2\theta - 24.7} \end{bmatrix}$$

$$B(x) = \begin{bmatrix} 0 \\ 0 \\ \dfrac{-1.8\cos\theta - 10.9}{\cos\theta - 24.7} \\ \dfrac{9.3\cos\theta + 38.6}{\cos^2\theta - 24.7} \end{bmatrix}. \tag{16}$$

In both systems, the pitch angle θ appears in the denominator of every non-zero entry of $B(x)$. Physically, this indicates that the effective inertia perceived by the actuator changes as the body tilts. This structural variation is precisely what the following mathematical test detects.

c) Certifying State-Dependent Contraction Rates: To apply Proposition 1 in practice, we evaluate the trace of $B(x)B(x)^\top$. If a system admits an orthogonal reduction to a uniform rate, this trace must remain strictly constant across the state space $\mathcal{X}$. For the CartPole (15), $\mathrm{tr}(B(x)B(x)^\top)$ varies monotonically by a factor of 4.9 across its domain. Similarly, the Segway model (16) exhibits a trace ranging from 3.368 to 4.372. Because this invariant scalar fluctuates in both systems, no orthogonal reduction $T(x)$ can exist. This mathematically confirms that both plants admit state-dependent contraction rates. Ultimately, as illustrated in Figure 2, the certified optimal rate $\lambda^*(x)$ varies by 0.977 for the CartPole and by 0.339 for the Segway.

d) Computation of the Contraction Rate Field: To compute the localized contraction rates (Figure 2), we evaluate a 15×15 grid over the two most active state dimensions for each plant using CV-STEM defined in Equations (9) to (12). The remaining state variables are marginalized by minimizing over 8 samples per grid cell, ensuring the mathematically required worst-case certification for the bounded subset. At each cell, the optimal rate λ^* is found via bisection on $\lambda \in [0.01, 60]$ (to a relative tolerance of 5×10^{-3}), selecting the highest λ for which the localized CV-STEM program remains feasible.

IV. State-Dependent IES for Optimal Policies

We analyze the incremental exponential stability (IES) of the RL-trained optimal policy defined in Equation (5), which accounts for the transient overshoot caused by prioritizing long-term optimality and state-dependent contraction rate.

We formally analyze the stability of the optimal policy defined in Equation (5) by establishing its state-dependent IES. Let $\|\delta x(t)\|_M^2 := \delta x^\top M(x)\delta x$ represent the Riemannian energy defined by CCMs along the geodesic joining $x(t)$ and $x^*(t)$. We define a reward function as follows:

$$R(x,\delta x) = -\|\delta x\|_M^2, \tag{17}$$

such that the optimal policy minimizes discounted cumulative Riemannian energy defined by CCMs. While [5] used normalized form of Equation (17) for numerical stability, we chose Equation (17) for analytical convenience.

Throughout the analysis, we assume the existence of a set of contracting policies.

Assumption 1. *Let $\{\mathcal{X}_n\}_{n=1}^N$ be a collection of nested, decreasing state subsets such that $\mathcal{X}_{n+1} \subset \mathcal{X}_n \subset \mathcal{X}_1 := \mathcal{X}$. For each $n = 1, 2, \ldots, N$, there exists a deterministic contracting policy $\hat{\pi}_n$ that satisfies Equation (2) with respect to $\|\delta x(t)\|_M^2$ at a rate λ_n as follows:*

$$\|\delta x(t_k)\|_M^2 \le \|\delta x(t_0)\|_M^2 e^{-2\lambda_n k\Delta t},\ \forall k\in\mathbb{N},\ \forall \delta x(t_0)\in\mathcal{X}_n, \tag{18}$$

where $0 < \lambda_1 < \lambda_2 < \cdots < \lambda_N < \infty$, and each $\mathcal{X}_n$ is forward invariant under its respective contracting policy $\hat{\pi}_n$.

We depict a notional diagram of Assumption 1 in Figure 1a.

1) State-Dependent IES: Let $\pi^\star$ denote the optimizer of Equation (5) with the reward function defined in Equation (17). Let $\mathcal{S}_n := \mathcal{X}_n \times \mathcal{X}_n \times \mathcal{U}$ be augmented RL state space for state space $\mathcal{X}_n$. We additionally define the state-dependent functions $\beta(s)$ and $\Lambda(s)$ used in the analysis as follows:

$$\Lambda(s) := \begin{cases} \lambda_N, & \text{if } s\in\mathcal{S}_N, \\ \lambda_n, & \text{if } s\in\mathcal{S}_n\setminus\mathcal{S}_{n+1}, \end{cases} \tag{19}$$

$$\beta(s) := \begin{cases} (1-\gamma e^{-2\lambda_N\Delta t})^{-1}, & \text{if } s\in\mathcal{S}_N, \\ (1-\gamma e^{-2\lambda_n\Delta t})^{-1}, & \text{if } s\in\mathcal{S}_n\setminus\mathcal{S}_{n+1}, \end{cases} \tag{20}$$

for $n = 1, 2, \ldots, N-1$.

Theorem 1 (State-Dependent IES)**.** *Suppose there exists a set of contracting policies $\{\hat{\pi}_n\}_n^N$ as in Assumption 1. Then, the optimal policy $\pi^\star$ of Equation (5) is incrementally exponentially stable at a state-dependent rate as follows:*

$$C(s_k) \le \beta(s_0)C(s_0)e^{-2\overline{\lambda}(0,k)k\Delta t},\quad \forall k\in\mathbb{N},\ s_0\in\mathcal{S}, \tag{21}$$

where $\beta(s)$ is defined in Equation (20), $\overline{\lambda}(0,k) := k^{-1}\sum_{k'=0}^{k-1}\Lambda(s_{k'})$, and $\Lambda(s)$ is defined in Equation (19).

Proof. See Appendix A. □

Figure 1b illustrates a notional state-dependent contraction bound, which permits the potentially faster contraction rates established in Theorem 1.

Corollary 1 (Nominal IES)**.** *When Assumption 1 is restricted to a single subset ($N = 1$), Theorem 1 reduces to the nominal IES at the uniform rate λ_1. Furthermore, restricting $\gamma \to 0$ gives overshoot-free nominal contracting policy.*

Remark 1 (Advantages of Minimizing Error Defined by Contraction Metrics)**.** *An optimal policy that minimizes the Euclidean error (i.e., $\|\delta x\|_I^2$) does not guarantee IES under the framework of Theorem 1, unless the dynamics in Equation* (1) *admit I as contraction metrics.*

Proof. See Appendix B. □

Remark 2 (Comparison to IES Established in [5])**.** *[5] does not account for transient overshoot when proving the IES of the optimal policy $\pi^\star$, resulting in a conservative guarantee where IES holds only if the discount factor is below a certain threshold. In contrast, Theorem 1 holds for any discount factor $\gamma \in (0, 1)$ by accounting for the transient overshoot introduced by long-term optimality via $\beta(s_0)$.*

V. Toy Problems

In this section, we use two nonlinear polynomial dynamical systems, for which exact contraction metrics and our proposed optimal controls can be derived, to illustrate Theorem 1 by answering the following questions:

1) Does the optimal control policy satisfy the state-dependent contraction bounds established in Theorem 1?
2) Does the discount factor systematically correlate with the running mean contraction rate, overshoot, and cumulative error?

To answer these questions, we use a λ-ascent reference control, which we describe in Section V-A5, to generate a reference trajectory from an arbitrary initial condition to satisfy the forward-invariance contraction rate in Assumption 1. For comparison, we generate another reference trajectory from the same initial condition using a stabilizing control, which we describe in Section V-A4, to illustrate the system's behavior when the forward invariance does not hold.

A. Polynomial Dynamical Systems

We consider the two-dimensional state and one-dimensional control ($n = 2, m = 1$) Moore–Greitzer jet-engine model [6] and the Duffing double-well oscillator [7] and show their control-affine dynamics in Equation (22) and Equation (23), respectively.

1) Control-Affine Forms: **Moore–Greitzer jet-engine model.** The Galerkin-projected post-stall compressor model is given by:

$$\begin{aligned}\dot{x}_1 &= -x_2 - \tfrac{3}{2}x_1^2 - \tfrac{1}{2}x_1^3,\\ \dot{x}_2 &= 3x_1 + u,\end{aligned} \tag{22}$$

where x_1 is the mass flow, x_2 is the pressure rise, and the control input u represents the throttle setting. The control problem here is to stabilize the system at the origin, which is a non-hyperbolic equilibrium of the open-loop dynamics, so the drift alone does not stabilize it.

Duffing double-well oscillator. The normalized Duffing oscillator, with stiffness $\alpha = 1$, softening coefficient $\beta = \frac{1}{3}$, and damping $\delta = \frac{3}{10}$, is given by:

$$\begin{aligned}\dot{x}_1 &= x_2,\\ \dot{x}_2 &= -x_1 + \tfrac{1}{3}x_1^3 - \tfrac{3}{10}x_2 + u,\end{aligned} \tag{23}$$

where x_1 is the position, x_2 is the velocity, and the control input u represents the external force applied to the oscillator. The control problem here is again to stabilize the system at the origin, which is a stable focus of the open-loop dynamics but only locally: the cubic term places saddle equilibria at $(\pm\sqrt{3}, 0)$ inside the state space, so the drift alone does not stabilize it either.

2) MDP Setting: Each of systems above will be simulated in discrete time $dt = 0.05$ and $H = 100$ timesteps, so $T = H\Delta t = 5\,\text{s}$. The state space is $\mathcal{X} = [-1.5, 1.5] \times [-2.5, 2.5]$ for (22) and $\mathcal{X} = [-2.5, 2.5]^2$ for (23), and the control space is $\mathcal{U} = [-6, 6]$ and $\mathcal{U} = [-8, 8]$, respectively, each twice the box from which the reference control is drawn so that feedback always has authority to spare. The rationale behind these numbers is to choose a fine enough discretization to reduce the numerical integration error, and also to ensure the given tracking admits a 95% reduction of the initial Riemannian energy over the horizon. The latter holds because T exceeds $1.5/\min_x \lambda(x)$ on both domains, which is $3.06\,\text{s}$ for (22) and $1.45\,\text{s}$ for (23). We hold H fixed across every discount factor, so that the execution length remains a property of the task rather than of γ.

3) Existence of State-Dependent Contraction Rates: Both plants share the constant input matrix $B = [0 \quad 1]^\top$, so $\operatorname{tr}(BB^\top)$ is constant and Proposition 1 is decided by the drift Jacobian $A(x) = \partial f/\partial x$ alone. Applying the same trace test to A, the Moore–Greitzer model (22) gives $\operatorname{tr}(A(x)A(x)^\top) = (3x_1 + \frac{3}{2}x_1^2)^2 + 10$, which ranges from 10.000 to 72.016 over $\mathcal{X}$, while the Duffing oscillator (23) gives $\operatorname{tr}(A(x)A(x)^\top) = (x_1^2 - 1)^2 + 1.09$, ranging from 1.090 to 28.652. Because this invariant scalar fluctuates in both systems, no orthogonal reduction $T(x)$ can exist, which confirms that both plants admit state-dependent contraction rates. The certified rate fields of Appendix F realize this, varying by $4.98\times$ for (22) and $1.84\times$ for (23).

4) Stabilizing Reference Control: Since neither drift is stabilizing over its domain, the reference trajectory of (51) cannot be generated with $u^* = 0$. For the Moore–Greitzer model, we apply the stabilizing control law $u^*(x) = -x_2$ to generate a reference trajectory for an arbitrary (unstable) initial condition, while for the Duffing oscillator, we utilize $u^*(x) = -2x_1$, which is the cancellation $u = -(\alpha + k_1)x_1$ with $k_1 = 1$ and needs no derivative term because the plant already carries the damping δ. Both closed loops are Hurwitz at the origin, with eigenvalues $-0.5 \pm 1.658\jmath$ and $-0.15 \pm 1.726\jmath$ respectively, and are certified asymptotically stable over the entire $\mathcal{X}$ by a quartic Lyapunov function, while no quadratic certificate exists for either. We detail this in Appendix F.

5) λ-Ascent Reference Control: The stabilizing reference stays where it starts in the rate field, so it never exercises the ordering of Assumption 1. To exercise it, we instead choose u^* to climb the rate field, differentiating λ along (51) and selecting the admissible input that maximizes $\dot{\lambda}$. The reference then begins in a low-rate region and migrates to a high-rate one while remaining a trajectory of the plant. We describe the law and the states at which a rate can actually be held in Appendix E.

B. Contraction Metric Synthesis using SoS Programming

We follow [8] that imposes the contraction condition as an algebraic identity rather than at sampled states. A polynomial p is nonnegative on the box $\mathcal{X} = \prod_i [\underline{x}_i, \overline{x}_i]$ whenever it admits the representation $p = \sigma_0 + \sum_i \sigma_i b_i$ with every σ_j a sum of squares (SoS) and $b_i(x) := (x_i - \underline{x}_i)(\overline{x}_i - x_i)$ [9], and this representation is itself found by semidefinite programming [10]. Writing the matrix inequality (42) as a scalar polynomial in an auxiliary variable $v \in \mathbb{R}^n$ and certifying each resulting polynomial in this form, we solve the following optimization:

$$\begin{aligned} \max_{\lambda,\, W,\, \{\sigma_j\}} \quad & \lambda \\ \text{s.t.} \quad & -v^\top S(x) v = \sigma_0 + \textstyle\sum_i \sigma_i b_i, \\ & v^\top (W(x) - \underline{w} I) v = \sigma_0' + \textstyle\sum_i \sigma_i' b_i, \\ & v^\top (\overline{w} I - W(x)) v = \sigma_0'' + \textstyle\sum_i \sigma_i'' b_i, \\ & \sigma_j,\, \sigma_j',\, \sigma_j'' \text{ SoS}, \end{aligned} \tag{24}$$

where $S := \dot{W} + AW + WA^\top + 2\lambda W - \frac{2}{r} BB^\top$ stands for the CV-STEM operator of (42) with $A := \partial f / \partial x$ and $\dot{W} = \sum_i (\partial W / \partial x_i) f_i$, and $\underline{w}, \overline{w}$ stand for the bounds on the spectrum of the dual metric $W = M^{-1}$. We take each entry of W of total degree 2, with $\underline{w} = 0.1$, $\overline{w} = 10$, and $r = 0.1$. Since λ multiplies W, the program is bilinear and we solve it by bisection on λ, each trial being a single linear SDP.

Intuitively, the output contraction metric is analytical expressions in terms of state x whose condition number over $\mathcal{X}$ is bounded by 27.7 for (22) and 34.0 for (23), and is exact under the criterion that the certified inequality holds at every point of $\mathcal{X}$ rather than at a finite draw from it. Note that this differs from contraction metric synthesis approach CV-STEM (9)–(12), which we will use for non-polynomial nonlinear systems in Section III-B, and the reason is CV-STEM is sample-based and does not provide analytical expressions of contraction metrics but in sample wise to reduce any approximation and estimation errors. The certified condition is identical in both, so the rates the two families report remain directly comparable, and we compare them in Appendix D.

C. Optimal Control Synthesis using SoS Programming

Given the certified metric, the benchmark optimal control is the global minimizer of the discounted tracking cost over the same discrete-time plant:

$$\begin{aligned} \min_{u_{0:H-1}} \quad & \textstyle\sum_{t=0}^{H-1} \gamma^t c(x_t, u_t) \\ \text{s.t.} \quad & x_{t+1} = x_t + \Delta t\, (f(x_t) + B u_t), \\ & x_t \in \mathcal{X}, \quad u_t \in \mathcal{U}, \end{aligned} \tag{25}$$

where x_0 is fixed and c stands for the negated reward of (17), written in the potential-shaped form $c(x_t, u_t) = -q(V_t - V_{t+1}) + r\|u_t\|^2$ with $V_t := \|x_t - x_t^*\|^2_{M(x_t)}$. This is a nonconvex polynomial optimization problem, for which the moment-SoS hierarchy [11] produces a monotone sequence of certified lower bounds.

A dense hierarchy is not viable here, since the number of nonconvex variables is $3H$ and therefore grows with the horizon; a discount sweep would then return certified optima at low γ and timeouts at high γ. We instead exploit the chain structure of (25), in which x_{t+1} couples only to (x_t, u_t), through correlative sparsity [12], [13]. This decomposes the relaxation into cliques whose size is independent of H, so that the cost grows linearly in the horizon.

Two reformulations expose that structure. First, the decrement cost telescopes into a positive weighted sum of the levels V_t, which removes the cross terms and places every objective monomial on a single clique. Second, the certificate is written in W, so $M = W^{-1}$ is rational; we lift each level to an epigraph variable v_t constrained by $v_t \det W(x_t) \geq (x_t - x_t^*)^\top \operatorname{adj} W(x_t)(x_t - x_t^*)$, which is polynomial and tight at the optimum since $\det W > 0$ on $\mathcal{X}$ and the weights are positive. The resulting cliques are $\{x_t, u_t, x_{t+1}\}$ and $\{x_t, v_t\}$, ordered so that the running-intersection property holds, and we relax them at orders 3 and 4 respectively.

The relaxation returns a certified lower bound on (25), while its first-order moments return a candidate input sequence whose rollout through the environment gives a feasible upper bound. The two bracket the global optimum, so every optimal control reported below carries a certificate of how far it can be from optimal.

HOW TO ANSWER EACH QUESTIONS

VI. Conclusion

In this paper, we proposed a theoretical condition to determine whether control-affine systems exhibit state-dependent contraction rates. Building on this, we established state-dependent contraction bounds for optimal policies trained via reinforcement learning (RL) to minimize cumulative tracking errors. Crucially, we demonstrated that through RL policy optimization, these bounds decay at the highest feasible contraction rate for any given state, even when the underlying contraction metric relies on a conservative, uniform rate. Future work will focus on rigorous empirical validation of these theoretical guarantees, specifically verifying that accelerated error decay is actively realized in regions of the state space that admit higher contraction rates.

Appendix A
Proof of Theorem 1

We introduce the following two lemmas to prove Theorem 1.

Lemma 1 (Sandwiched Value Function)**.** *Suppose there exists a set of contracting policies $\{\hat{\pi}_n\}_n^N$ as in Assumption 1. Then, the cost value function for the optimal policy $\pi^\star$ of Equation* (5) *satisfies the following inequality:*

$$C(s) \leq V_C^{\pi^\star}(s) \leq \beta(s) C(s), \tag{26}$$

where $\beta(s)$ is defined in Equation (20)*.*

Proof. The lower bound holds trivially for $C \geq 0$, as the value function is the sum of non-negative costs. The upper bound follows from the optimality of $\pi^\star$, yielding

$$V_C^{\pi^\star}(s) \leq V_C^{\pi_{c^*}}(s) \leq \frac{C(s)}{1-\gamma e^{-2\lambda_N \Delta t}}, \quad \forall s \in \mathcal{S}_N, \tag{27}$$

$$V_C^{\pi^\star}(s) \leq V_C^{\pi_c}(s) \leq \frac{C(s)}{1-\gamma e^{-2\lambda_n \Delta t}}, \quad \forall s \in \mathcal{S}_n \setminus \mathcal{S}_{n+1}, \tag{28}$$

where we use the convergence of a geometric series. □

Lemma 2 (Value Contraction)**.** *Suppose there exists a set of contracting policies $\{\hat{\pi}_n\}_n^N$ as in Assumption 1. Let $\Lambda(s)$ be defined as in Equation* (19)*. Then, the cost value function for the optimal policy $\pi^\star$ of Equation* (5) *achieves contraction at a running mean rate as follows:*

$$V_C^{\pi^\star}(s_k) \leq V_C^{\pi^\star}(s_0) e^{-2\overline{\lambda}(0,k)k\Delta t}, \quad \forall k \in \mathbb{N},\ s_0 \in \mathcal{S}, \tag{29}$$

where $\overline{\lambda}(0,k) := k^{-1}\sum_{k'=0}^{k-1}\Lambda(s_{k'})$ is the running average contraction rate along the trajectory.

Proof. Suppose the cost C is nonzero, so is $V_C^{\pi^\star}$. Given the deterministic dynamics and the deterministic optimal policy, we begin with the Bellman optimality equation:

$$V_C^{\pi^\star}(s_t) = C(s_t) + \gamma V_C^{\pi^\star}(s_{t+1}), \tag{30}$$

which can be rearranged to isolate the contraction factor:

$$V_C^{\pi^\star}(s_{t+1}) = \underbrace{\gamma^{-1}\left(1 - \frac{C(s_t)}{V_C^{\pi^\star}(s_t)}\right)}_{\rho(s_t)} V_C^{\pi^\star}(s_t). \tag{31}$$

Dividing the upper bounds in Lemma 1 by $V_C^{\pi^\star}(s_t) > 0$ yields:

$$1 \leq \beta(s_t)\frac{C(s_t)}{V_C^{\pi^\star}(s_t)} \implies \beta(s_t)^{-1} \leq \frac{C(s_t)}{V_C^{\pi^\star}(s_t)} \leq 1,\ \forall s_t \in \mathcal{S}. \tag{32}$$

Rearranging Equation (32) and scaling by γ^{-1} establishes the following uniform bounds on $\rho(s_t)$:

$$\rho(s_t) \leq \gamma^{-1}\big(1-\beta(s_t)^{-1}\big) = e^{-2\Lambda(s_t)\Delta t} < 1, \quad \forall s_t \in \mathcal{S}, \tag{33}$$

which yields the step-wise value contraction:

$$V_C^{\pi^\star}(s_{t+1}) \leq e^{-2\Lambda(s_t)\Delta t} V_C^{\pi^\star}(s_t). \tag{34}$$

Recursively applying the value contraction over k steps gives:

$$V_C^{\pi^\star}(s_{t+k}) \leq V_C^{\pi^\star}(s_t)\prod_{k'=0}^{k-1} e^{-2\Lambda(s_{t+k'})\Delta t} \tag{35}$$

$$= V_C^{\pi^\star}(s_t) e^{-2\overline{\lambda}(t,t+k)k\Delta t}, \tag{36}$$

where $\overline{\lambda}(t,t+k) := k^{-1}\sum_{k'=0}^{k-1}\Lambda(s_{t+k'})$ denotes the running average contraction rate along the trajectory. □

Proof of Theorem 1*.* Applying the value sandwich inequality (see Lemma 1) to the value contraction inequality (see Lemma 2 gives the desired results as follows:

$$V_C^{\pi^\star}(s_{t+k}) \leq V_C^{\pi^\star}(s_t) e^{-2\overline{\lambda}(t,t+k)k\Delta t} \tag{37}$$

$$\implies C(s_{t+k}) \leq \beta(s_t) C(s_t) e^{-2\overline{\lambda}(t,t+k)k\Delta t}, \tag{38}$$

for all $k \in \mathbb{N}$ and $s_t \in \mathcal{S}$. Setting $t = 0$ gives Equation (21). □

Appendix B
Proof of Remark 1

The optimal policy minimizing Euclidean error satisfies Lemma 1 with an additional slack term, $\sqrt{\overline{m}/\underline{m}}$, as the contraction bound of policy $\hat{\pi}_n$ in Euclidean space becomes:

$$C(s_k) \leq \sqrt{\overline{m}/\underline{m}}\, C(s_0) e^{-\lambda k \Delta t}, \quad \forall k \in \mathbb{N},\ s_0 \in \mathcal{S}. \tag{39}$$

This renders Equation (32) invalid; consequently, the value contraction in Lemma 2 is no longer guaranteed.

Appendix C
Computation of the Contraction-Rate Field

We detail the procedure summarized in Section III-B and the conditions under which it is valid. Throughout, $\mathcal{Q}$ denotes the finite set of query states at which the field is evaluated.

a) Two Estimators: The map $x \mapsto \lambda^*(x)$ of (6) admits two distinct finite-sample surrogates, and they answer different questions:

- *Certified.* Solve (9)–(12) *once* over all of $\mathcal{Q}$ at the prescribed uniform rate, then report the local rate of the resulting single metric field at each query state,

$$\lambda(x) := -\tfrac{1}{2}\,\lambda_{\max}\big(\widehat{MA_{\mathrm{cl}}},\, M\big), \quad A_{\mathrm{cl}} := A - \tfrac{1}{r}BB^\top M, \tag{40}$$

  where $\lambda_{\max}(\cdot,\cdot)$ denotes the largest generalized eigenvalue and the arguments are omitted for brevity. This is the rate of the metric that is actually deployed.
- *Ceiling.* Re-solve the program independently at each query state and report $\lambda^*(\{x\})$ by search over λ. This is the rate attainable if the metric were redesigned for that state alone; it upper bounds the certified rate and is not realized by any single controller.

Both are invariant under the orthogonal congruence of Proposition 1, hence exactly constant on uniform-rate plants. Figure 2 reports the certified estimator. If the joint program is infeasible at the prescribed rate, this is a statement about the certified envelope of the plant and not a failure of the visualization; it must be reported as such rather than worked around by lowering λ.

b) Marginalization: Two of the n dimensions are gridded and the remainder are marginalized by minimization, which is the worst-case certification required for a bounded subset: reporting an average or a single slice would claim a rate that does not hold on part of the cell. Only *active* dimensions—those on which $A(\cdot)$ and $B(\cdot)$ actually depend—are sampled; inactive dimensions, in particular the translation directions of $\mathcal{X}$, are pinned at the box center. Randomizing an inactive dimension inflates the sample count without adding a single distinct constraint, and any spread it appears to produce is solver noise.

c) Actuator Feasibility: A certificate is admissible only if the induced feedback respects $\mathcal{U}$, so each cell is additionally screened on

$$\mathrm{Pr}_{\delta x, u^*}\big[\, u^* - \tfrac{1}{r}B^\top M\,\delta x \notin \mathcal{U}\,\big] \leq \eta, \qquad \eta = 0.05, \tag{41}$$

estimated by Monte Carlo. The number of draws is not a free parameter. The gate is evaluated one state at a time, so the draw count *is* the entire sample: at 64 draws the estimator's standard deviation is 0.023 against a budget of $\eta = 0.05$, and for a plant whose true violation rate sits 0.65σ below the budget this rejects 25.7% of cells at random—17.3% observed on the car. Because a rejected cell falls back to a lower reported rate, the coin flip is drawn directly into the figure as spatial structure. We use 2×10^3 draws per cell.

A residual ripple in (41) survives any draw count and is not noise. Constructing $W(\psi) = T(\psi)W_0T(\psi)^\top$ from the exact symmetry of Proposition 1—no solver in the loop, so the true rate is constant by construction—still yields a violation rate ranging over 0.02585–0.02704, a peak-to-peak spread of 3.3σ at 2×10^5 draws. The cause is a frame mismatch: δx is drawn from a fixed box in world coordinates while $K(x) = r^{-1}B^\top M$ co-rotates with the state, so the gate measures the box, not the plant. The draw box must be expressed in the same frame as the gain.

d) Resolution: Searching λ on a fixed geometric ladder $\lambda \in [0.01, 60]$ of L points reports only ladder values, so the coarsest spread the estimator can express is one bin, $(60/0.01)^{1/(L-1)}$. At $L = 24$ this equals 1.4597, and every spread we initially measured was an integer power of it (1.46, 2.131, 3.110, 20.613), with as few as two distinct values across 225 cells: the reported state dependence was the ladder. Feasibility in λ is non-monotone—χ carries weight c_0, so the realizable set is an interval rather than a ray—which is why the ladder is swept rather than bisected. We therefore keep the sweep to bracket the interval and then bisect only between the last feasible ladder point and its successor, to a relative tolerance of 5×10^{-3}, at a cost of 24–40 solves per cell.

e) Declaring a Field Constant: Whether a field is constant is a scale-free question and must be tested as one. An absolute tolerance fails: the car's certified field spans 1.87×10^{-6} in absolute terms, which is 4.2×10^{-6} *relative* to its mean rate of 0.442, yet exceeds a 10^{-6} absolute threshold and is then rendered on an autoscaled colormap—painting a plant that is uniform to six digits as a rate gradient. We declare a field constant when its peak-to-peak spread is at most 10^{-3} of its mean, and render it at a fixed level.

f) Verification: The uniform-rate classification of the car and the quadrotor is confirmed by three independent computations: with (41) disabled, the ceiling estimator returns $\lambda^* = 5.03358$ at all 15 heading values with a spread of 1.000000; the certified estimator spans 1.87×10^{-6} and 2.11×10^{-7} respectively; and (40) evaluated over 2×10^3 and 3×10^3 uniform samples of $\mathcal{X}$ gives a ratio of 1.0000. For the two state-dependent plants of Section III-B the certified spreads are 3.503 (Segway) and 2.217 (CartPole).

Appendix D
Certification on Polynomial Plants: Sum-of-Squares vs. Sampled CV-STEM

Both plant families are certified against the same condition on the dual metric $W = M^{-1}$,

$$S = \dot{W} + AW + WA^\top + 2\lambda W - \tfrac{2\nu}{r}BB^\top \preceq 0, \qquad \dot{W} = \sum_{i=1}^{n} \frac{\partial W}{\partial x_i} f_i(x), \tag{42}$$

with $A = \partial f/\partial x$. For a general nonlinear f this is imposed at N drawn states, which is the program of (9)–(12), repeated here:

$$\min_{\bar{W}_k,\, \nu,\, \chi} \ c_0\chi + c_1\nu \tag{43}$$

$$\text{s.t. } I \preceq \bar{W}_k \preceq \chi I, \quad S_k \preceq -\varepsilon I, \quad \forall k \in \{1, \ldots, N\}, \tag{44}$$

$$\nu > 0, \quad \nu \le \underline{w}^{-1}, \quad \chi \le \nu \overline{w}, \tag{45}$$

S_k being (42) evaluated at x_k. When f and B are polynomial the same condition can instead be certified on the whole box $\mathcal{X} = \prod_i [\underline{x}_i, \overline{x}_i]$. Writing $W(x) = \sum_{|\alpha| \le d} C_\alpha x^\alpha$ with $C_\alpha \in \mathbb{S}^n$ unknown makes $S(x)$ a polynomial matrix, which is scalarized with a fresh $v \in \mathbb{R}^n$ and discharged by Putinar's Positivstellensatz on the box generators $g_i(x) \coloneqq (x_i - \underline{x}_i)(\overline{x}_i - x_i)$:

$$\text{find } \ \{C_\alpha\}, \ \{\sigma_j, \tau_j, \rho_j\} \tag{46}$$

$$\text{s.t. } \ -v^\top S(x) v = \sigma_0 + \textstyle\sum_i \sigma_i\, g_i, \tag{47}$$

$$v^\top (W(x) - \underline{w} I) v = \tau_0 + \textstyle\sum_i \tau_i\, g_i, \tag{48}$$

$$v^\top (\overline{w} I - W(x)) v = \rho_0 + \textstyle\sum_i \rho_i\, g_i, \tag{49}$$

$$\sigma_j, \tau_j, \rho_j \text{ SOS in } (x, v), \text{ quadratic in } v. \tag{50}$$

In both programs λ multiplies the unknown metric, so each is a linear SDP at fixed λ and the rate follows by bisection. The four subsections below isolate what changes between them.

A. Quantifier: N samples vs. the whole box

(44) is N separate matrix inequalities and its solution is N unrelated matrices $\bar{W}_k$; a metric *field* is recovered only afterwards, by regressing a network onto them. The certificate holds at the N queried states, and away from them rests on the regression residual, which is why the sampled estimator is reported together with a verification sweep. By contrast (47) is one polynomial identity: a single finite SDP discharges the uncountable family $\{S(x) \preceq 0 : x \in \mathcal{X}\}$, and its output is an analytic $W(x)$, hence an $M(x) = W(x)^{-1}$ that is rational in x and exact at every state.

B. The derivative $\dot{W}$: finite differences vs. exact

$\bar{W}$ in (44) is a table of numbers, not a function, so $\dot{\bar{W}}_k$ has to be finite differenced along stored trajectories — a second approximation, on top of the sampling, sitting inside the certificate. In the polynomial case $\dot{W} = \sum_i (\partial W/\partial x_i) f_i(x)$ is formed symbolically from the same coefficients $\{C_\alpha\}$ that are being solved for, so it is exact and needs no trajectory data at all.

C. Normalization: (ν, χ) vs. certified bounds

(44)–(45) pin the metric scale indirectly, through the normalization $\bar{W} = \nu W$ and the variables ν, χ, because $I \preceq \bar{W}_k \preceq \chi I$ constrains only the sampled matrices. (48)–(49) bound W at every x directly, so ν and χ disappear as variables, $\underline{w}$ and $\overline{w}$ enter as constants, and (42) is used at $\nu = 1$. The SOS program is consequently pure feasibility at each λ: there is no $c_0\chi + c_1\nu$ objective, hence no trade-off weight to tune.

D. Problem size and applicability

The sampled program has $O(N)$ PSD blocks of size n, so its cost is set by the sample count and it applies to any f. The SOS program's unknowns are $\{C_\alpha\}$ plus one Gram matrix per multiplier, and its constraints are coefficient-matching equalities, one per monomial in (x, v) — the count is set by the degree, not by a sample count. The restriction of σ_j, τ_j, ρ_j to degree two in v in (50) is what makes the right-hand sides reproduce $v^\top(\cdot)v$ exactly, with no spurious v^0 or v^4 terms. The price is that f and B must be polynomial and that each Gram block has side $n\binom{n+\lceil\delta/2\rceil-1}{\lceil\delta/2\rceil}$ for a multiplier of degree δ in x, so the construction is practical for the low-dimensional polynomial plants and not for the full classic and Isaac families. Sampled CV-STEM is the general-purpose instrument; SOS is the exact one where the plant admits it. Because both certify (42) with the same r, $\underline{w}$, $\overline{w}$ and sign convention, the rates they report are the same quantity and remain directly comparable.

Appendix E
Reference Trajectories with Ascending Contraction Rate

Assumption 1 orders the subsets $\mathcal{X}_n$ by increasing λ_n, so exercising the state-dependent bound of Equation (21) requires a reference that begins in a low-rate region and migrates to a high-rate one. The reference is not free to be placed pointwise: it must satisfy (1) exactly, so the only design variable is $u^* \in \mathcal{U}$ in

$$x^*_{t+1} = x^*_t + \Delta t\left(f(x^*_t) + B(x^*_t)u^*_t\right). \tag{51}$$

a) Ascent by Inverse Dynamics: Differentiating the local rate (40) along (51) separates the two contributions,

$$\dot{\lambda} = \nabla\lambda^\top f + \left(B^\top\nabla\lambda\right)^\top u^*, \tag{52}$$

in which only the second term is actuated. Maximizing it over a ball gives $u^* \propto B^\top\nabla\lambda$, but this alone is insufficient: the drift term is sign-indefinite and dominates on an unstable plant, and applying the ascent direction by itself drove λ from 0.1232 to -0.6885 on the Segway even though its ascent contribution is nonnegative by construction. We therefore cancel the drift by inverse dynamics before ascending,

$$u^*_t = \Pi_{\mathcal{U}}\Big(\underbrace{B^\dagger\left(-f(x^*_t)\right)}_{\text{drift cancellation}} + k\,\underbrace{B^\top\nabla\lambda(x^*_t)}_{\text{steepest ascent}}\Big), \tag{53}$$

with $\Pi_{\mathcal{U}}$ the projection onto the input box, which yields $\dot{\lambda} \approx k\|B^\top\nabla\lambda\|^2 \geq 0$ whenever the box is inactive. Using the pseudoinverse makes the law underactuation-safe: it realizes the best available projection of the requested direction, and if $\nabla\lambda$ lies entirely outside $\mathrm{range}(B)$ then $B^\top\nabla\lambda = 0$ and the law reports that immediately rather than issuing a silent no-op. We take $\nabla\lambda$ by central differences with step $h = h_0(x_{\max} - x_{\min})$, since (40) already contains a Jacobian and a generalized eigendecomposition and an analytic gradient would require second derivatives of f. Components of $\nabla\lambda$ pointing out of $\mathcal{X}$ within a margin of a face are zeroed, so the ascent runs along the boundary instead of through it; otherwise the reference leaves $\mathcal{X}$ and the metric is evaluated where it was never certified.

b) When the Construction Applies: (53) requires two properties, and both must be verified before use. First, the rate must genuinely vary, per Appendix C. Second—and this is the binding condition—the low-rate region must be one in which a reference can *dwell* for a useful fraction of the horizon. The two are independent, and the second fails on the Segway (16) even though the first holds:

- Its low-rate region is the large-$|\theta|$ set, and θ is the only coordinate with any rate dependence ($\mathrm{corr}(\lambda, |\theta|) = -0.226$).
- Holding a tilt is a relative equilibrium: with $\omega = 0$ and v free, the last two rows of $f + Bu = 0$ are linear in (v, u) and solve exactly as $v^* \approx -181\,\theta$, $u^* \approx -940\,\theta$. At $\theta = 0.05$ this demands $v^* = -9.06$ against $|v| \leq 1$ and $u^* = -56.97$ against $|u| \leq 24$, so only $|\theta| < 0.006$ is admissible.
- Consequently the reference leaves $\mathcal{X}$ within 9.93, 1.74 and 0.66 seconds of a 60-second horizon from $\theta_0 = 0.05$, 0.30 and 0.85, always through the velocity bound and at a peak $|u|$ of 6.5–9.7 of the 24 available: the binding constraint is $\mathcal{X}$, not $\mathcal{U}$. Under a greedy input maximizing the distance to the faces of $\mathcal{X}$, the only initial condition that survives the full horizon sustains a mean tilt of 0.0000, which is the maximum of λ.

On the Segway, therefore, low rate and long dwell are mutually exclusive: the rate is low exactly where the plant is falling (unstable eigenvalue $+2.76$, e-folding time 0.36 s), and a falling state is not a reference. Every candidate reference initialization we evaluated under (53)—centered, mid-tilt, maximum tilt, and both (θ, ω) diagonals—pins to a face of $\mathcal{X}$ within a few seconds and λ *decreases* by a factor of 0.26–0.70. Where both properties hold, the construction behaves as designed. On the kinematic car restricted to a low-speed velocity box—where the Hautus margin $\min(1, v)$ makes the rate state-dependent, while the drift acts only along the translation directions, so $u^* = 0$ holds every state of the active subspace exactly and dwell is never binding—the certified field spans 5.635 and (53) raises λ by $1.401\times$ along the reference with the input box never active.

Appendix F
Reference Controls and Contraction Metrics for the Toy Problems

This appendix certifies the two ingredients Section V takes for granted: that the reference controls u^* stabilize their plants, and that the metrics used to measure the Riemannian energy are contraction metrics on the stated boxes.

A. The Reference Controls Are Stabilizing

Neither drift is stable, so the reference of (51) cannot be generated with $u^* = 0$. We use

$$u^*_{\mathrm{mg}}(x) = -x_2, \qquad u^*_{\mathrm{duff}}(x) = -2x_1, \tag{54}$$

which give the closed loops

$$\begin{aligned} g_{\mathrm{mg}}(x) &= \left(-x_2 - \tfrac{3}{2}x_1^2 - \tfrac{1}{2}x_1^3,\ \ 3x_1 - x_2\right), \\ g_{\mathrm{duff}}(x) &= \left(x_2,\ \ -3x_1 + \tfrac{1}{3}x_1^3 - \tfrac{3}{10}x_2\right). \end{aligned} \tag{55}$$

Both are Hurwitz at the origin: the Jacobians are $[\,0\ \ {-1};\ 3\ \ {-1}\,]$ with eigenvalues $-\frac{1}{2}\pm\frac{\sqrt{11}}{2}\jmath=-0.5\pm1.658\jmath$, and $[\,0\ \ 1;\ {-3}\ \ {-\frac{3}{10}}\,]$ with eigenvalues $-0.15\pm1.726\jmath$. The sign matters and is easy to get backwards: the opposite choice $u=+2x_1$ on (23) yields eigenvalues $\{+0.861,\,-1.161\}$, a saddle.

Local stability is not enough, since the reference must remain well behaved over the whole box. We therefore certify each closed loop on its own $\mathcal{X}$ by sum of squares, seeking V with

$$V-\varepsilon\|x\|^2\succeq_{\mathrm{SOS}}0,\quad -\nabla V^\top g-\varepsilon\|x\|^2\succeq_{\mathrm{SOS}}0\quad\text{on }\mathcal{X},\tag{56}$$

each certificate carrying its own multipliers for the box faces $(x_i-\underline{x}_i)(\overline{x}_i-x_i)$, with $\varepsilon=10^{-3}$ and V normalized by fixing the coefficient of x_1^2 to 1.

Proposition 2 (Stabilization on $\mathcal{X}$)**.** *For* (55) *the programs* (56) *are infeasible at* $\deg V=2$ *and feasible at* $\deg V=4$*, with*

$$\begin{aligned}V_{\mathrm{mg}}(x)&=x_1^2+0.3600\,x_2^2-0.3286\,x_1x_2-0.2735\,x_1^3+0.6541\,x_1^2x_2\\&\quad-0.2234\,x_1x_2^2+0.1581\,x_2^3+0.0608\,x_1^4-0.1519\,x_1^3x_2\\&\quad+0.1843\,x_1^2x_2^2-0.0715\,x_1x_2^3+0.0333\,x_2^4,\\V_{\mathrm{duff}}(x)&=x_1^2+0.3364\,x_2^2+0.0404\,x_1x_2-0.0443\,x_1^4+0.0206\,x_1^3x_2\\&\quad+0.0268\,x_1^2x_2^2+0.0191\,x_1x_2^3+0.0052\,x_2^4.\end{aligned}$$

Hence the origin is asymptotically stable with $\mathcal{X}$ contained in its region of attraction, and every reference generated by (51) *under* (54) *stays bounded on $\mathcal{X}$.*

That no *quadratic* V certifies either box is worth stating rather than hiding: on both plants the cubic term is what carries the state dependence, and a quadratic certificate would be claiming a rate whose validity is exactly the question. Verified independently of the solver on a 601^2 grid over $\mathcal{X}$, the printed (rounded) certificates satisfy

	$\min V/\|x\|^2$	$\max V/\|x\|^2$	$\max \nabla V^\top g/\|x\|^2$
(22)	0.1375	2.5949	-0.0492
(23)	0.3360	1.0000	-0.0690

so V is positive definite and $\dot V$ negative definite on $\mathcal{X}\setminus\{0\}$. (The exponential rate implied by these constants, $\dot V\le-(c_2/c_3)V$, is very conservative and is not the rate reported anywhere else in this paper; it certifies stabilization only.)

B. Certified Contraction Metrics

The metrics are obtained from the same condition used throughout, imposed as an algebraic identity rather than at sampled states. For a polynomial plant, a polynomial p is nonnegative on a box whenever $p=\sigma_0+\sum_i\sigma_i(x_i-\underline{x}_i)(\overline{x}_i-x_i)$ with every σ_j SOS, which holds at *every* point of the box and not at a draw from it. We take $W(x)$ with entries of total degree 2 and impose

$$-\dot W+AW+WA^\top-\tfrac{2}{r}BB^\top+2\lambda W\preceq0,\quad \underline{w}I\preceq W(x)\preceq\overline{w}I,\tag{57}$$

on $\mathcal{X}$, with $\dot W=\sum_i(\partial W/\partial x_i)f_i$, $\underline{w}=0.1$, $\overline{w}=10$ and $r=0.1$. Since λ multiplies W, the largest certifiable rate is found by bisection, each trial being one linear SDP. This is the CV-STEM condition of (9)–(12), not the $B_\perp$-projected CCM condition: the latter drops the r term, so its λ would certify that *some* contracting controller exists while the deployed feedback is the Riccati form $u=-\frac{1}{r}B^\top M\delta x$, and the certified and the measured rate would then be different quantities.

The bisection gives ceilings of $\lambda=1.7188$ for (22) and $\lambda=2.8516$ for (23). We deploy 10% of each, $\lambda=0.1719$ and $\lambda=0.2852$, and this is deliberate. Maximizing the uniform rate *flattens* $\lambda(x)$: the optimizer equalizes it across the box, and a state-dependent plant certified into a near-constant rate field is precisely what these examples exist not to be. Measured on a 101^2 grid for (22), the spread $\max_x\lambda(x)/\min_x\lambda(x)$ runs 2.510, 3.232, 3.937, 4.975 at $\lambda=1.7188$, 0.8594, 0.4297, 0.1719 respectively.

The resulting certificates, with W symmetric and $M=W^{-1}$, are

$$\begin{aligned}W_{11}^{\mathrm{mg}}&=0.7991+0.2064\,x_1-0.0935\,x_2+0.1987\,x_1^2\\&\quad-0.0978\,x_1x_2+0.0132\,x_2^2,\\W_{12}^{\mathrm{mg}}&=1.4753-0.3299\,x_1-0.0896\,x_2-0.0744\,x_1^2\\&\quad-0.0257\,x_1x_2-0.0234\,x_2^2,\\W_{22}^{\mathrm{mg}}&=5.1170-0.4997\,x_1-0.3038\,x_2-0.0371\,x_1^2\\&\quad+0.0253\,x_1x_2-0.0288\,x_2^2,\end{aligned}$$

and, for the Duffing oscillator, a field even in x (its plant is odd, so no linear terms survive),

$$\begin{aligned}W_{11}^{\mathrm{duff}}&=1.0379+0.0482\,x_1^2+0.0666\,x_1x_2+0.0397\,x_2^2,\\W_{12}^{\mathrm{duff}}&=-1.9431-0.0938\,x_1^2-0.0611\,x_1x_2-0.0259\,x_2^2,\\W_{22}^{\mathrm{duff}}&=5.2872+0.1134\,x_1^2+0.0014\,x_1x_2-0.0405\,x_2^2.\end{aligned}$$

Both verify on a 101^2 grid with the largest LMI eigenvalue of (57) strictly negative. Their properties on $\mathcal{X}$, evaluated on a 201^2 grid, are

	λ deployed	$\lambda(x)$ range	spread	$\kappa(W)$
(22)	0.1719	[0.489, 2.433]	4.98×	≤ 27.7
(23)	0.2852	[1.035, 1.899]	1.84×	≤ 34.0

with $\lambda(x)$ the certified pointwise estimator (40) of the deployed metric, and $\operatorname{eig}W\in[0.217,6.980]$ and $[0.201,7.477]$ respectively, comfortably inside $[\underline{w},\overline{w}]$. The gap between the deployed uniform λ and $\min_x\lambda(x)$ is the slack Assumption 1 formalizes and Theorem 1 allows an optimal policy to convert into faster decay.

C. Where the Rate Can Actually Be Held

Exercising Assumption 1 requires the reference to migrate from a low-rate region to a high-rate one, and the migration cannot target an arbitrary high-λ state: the reference must satisfy (51), so a state can be *held* only where $f(x)+Bu=0$ for some admissible u. For (22) this makes $x_1=c$ an equilibrium exactly when $x_2=-\frac{3}{2}c^2-\frac{1}{2}c^3$ and $u=-3c$, so $|u|\le3$ caps $|c|\le1$, and λ along that curve runs

$[0.554, 1.165]$—a $2.10\times$ span against the field's $4.98\times$. The faster part of the field is reachable only in passing, by any reference whatsoever. We therefore place the start and target bands at $c = -0.90$ ($\lambda = 0.571$) and $c = +0.90$ ($\lambda = 1.140$), which keeps 0.30 of the actuator in reserve for the migration itself; holding either endpoint $c = \pm 1$ would consume the entire input box and leave nothing to steer with. For (23) every $(x_1, 0)$ is an equilibrium, with $u = x_1 - \frac{1}{3}x_1^3$ well inside $[-4, 4]$, so the choice is not actuator-limited and the target is simply where λ is largest on that curve.

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